

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

5. Develop an AWT-based Employee Salary Calculator.

The application should accept:

Employee ID

Employee Name

Basic Salary

Calculate: HRA = 20% of Basic Salary

DA = 10% of Basic Salary

Gross Salary = Basic + HRA + DA- Use suitable AWT components and event handling."

Test Cases: tc1_input- Basic Salary=10000, Click Calculate tc1_output- HRA=2000,

DA=1000, Gross Salary=13000 tc2_input- Basic Salary=25000, Click Calculate

tc2_output- HRA=5000, DA=2500, Gross Salary=32500 tc3_input- Basic Salary=50000,

Click Calculate tc3_output- HRA=10000, DA=5000, Gross Salary=65000

CODE:

```
import java.awt.*; import
```

```
java.awt.event.*;
```

```
public class EmployeeSalaryCalculator extends Frame implements ActionListener {
```

```
    Label idLabel, nameLabel, basicLabel, hraLabel, daLabel, grossLabel;
```

```
    TextField idField, nameField, basicField, hraField, daField, grossField;
```

```
Button calculateButton, clearButton;
```

```
public EmployeeSalaryCalculator() {
```

```
    setTitle("Employee Salary Calculator");
```

```
    setSize(500, 400);    setLayout(new
```

```
    GridLayout(7, 2, 10, 10));
```

```
    idLabel = new Label("Employee ID:");
```

```
    idField = new TextField();
```

```
    nameLabel = new Label("Employee Name:");
```

```
    nameField = new TextField();
```

```
    basicLabel = new Label("Basic Salary:");
```

```
    basicField = new TextField();
```

```
    hraLabel = new Label("HRA (20%):");
```

```
    hraField = new TextField();
```

```
    hraField.setEditable(false);
```

```
    daLabel = new Label("DA (10%):");
```

```
    daField = new TextField();
```

```
    daField.setEditable(false);    grossLabel = new
```

```
Label("Gross Salary:");    grossField = new  
TextField();    grossField.setEditable(false);
```

```
    calculateButton = new Button("Calculate");  
clearButton = new Button("Clear");
```

```
    calculateButton.addActionListener(this);  
clearButton.addActionListener(this);
```

```
    add(idLabel);  
add(idField);
```

```
    add(nameLabel);  
add(nameField);
```

```
    add(basicLabel);  
add(basicField);
```

```
    add(hraLabel);  
add(hraField);
```

```
    add(daLabel);  
add(daField);
```

```
    add(grossLabel);  
add(grossField);
```

```
        add(calculateButton);
add(clearButton);

        addWindowListener(new WindowAdapter() {
public void windowClosing(WindowEvent e) {
            System.exit(0);
        }
});

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {

        if (e.getSource() == calculateButton) {
            try
        {
                double basicSalary = Double.parseDouble(basicField.getText());

                if (basicSalary < 0) {
hraField.setText("Invalid Salary");
daField.setText("Invalid Salary");
grossField.setText("Invalid Salary");                return;
                }
            }
        }
    }
}
```

```

        double hra = basicSalary * 0.20;

double da = basicSalary * 0.10;          double
grossSalary = basicSalary + hra + da;


        hraField.setText(String.valueOf((int) hra));

daField.setText(String.valueOf((int) da));          grossField.setText(String.valueOf((int)
grossSalary));


        } catch (NumberFormatException ex) {
hraField.setText("Invalid Input");
daField.setText("Invalid Input");
grossField.setText("Invalid Input");
        }
    }

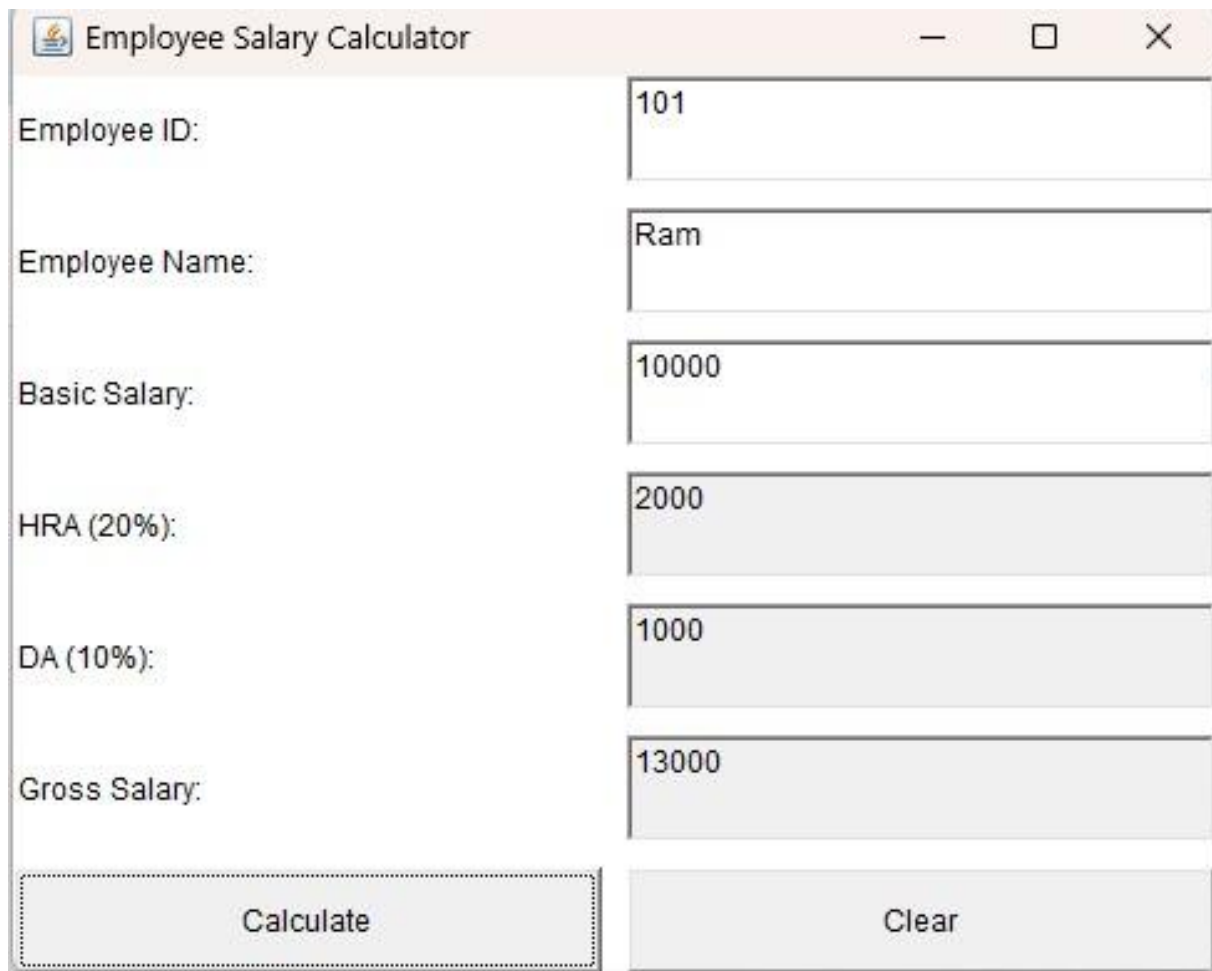
    if (e.getSource() == clearButton) {
idField.setText("");
nameField.setText("");
basicField.setText("");
hraField.setText("");          daField.setText("");
grossField.setText("");

    }
}

```

```
public static void main(String[] args) {  
    new EmployeeSalaryCalculator();  
}  
}
```

OUTPUT:



The screenshot shows a Java Swing window titled "Employee Salary Calculator". The window contains a form with the following fields and values:

Field	Value
Employee ID:	101
Employee Name:	Ram
Basic Salary:	10000
HRA (20%):	2000
DA (10%):	1000
Gross Salary:	13000

At the bottom of the window, there are two buttons: "Calculate" and "Clear".